FED-STD-H28/18A
10 May 1985
SUPERSEDING
FED-STD-H28/18
31 August 1978

FEDERAL STANDARD

SCREW-THREAD STANDARDS FOR FEDERAL SERVICES

SECTION 18

PHOTOGRAPHIC EQUIPMENT THREADS

This standard was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

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New and revised pages will be issued under Change Notices which will be numbered consecutively and will bear the date of issuance. Change Notices should be retained and filed in front of the standard until such times as they are superseded by a reissue of the entire standard.

FOREWORD

This section was developed in order to provide photographic equipment thread data for the Federal Services. The present issue is a complete revision of FED-STD-H28/18 dated 31 August 1978.

FED-STD-H28/18A was prepared by the Defense Industrial Supply Center (DLA-IS). The following significant changes from the previous issue are included:

(1) Connection details have been replaced by references to controlling standards.

(2) Threads for attaching mounted lenses, which originally appeared in the American National Standard ANSI PH3.10-1954, have been removed. The ANSI PH3.10-1954 was withdrawn as a standard as of 18 January 1965.

(3) Attachment thread sizes for lens accessories, which originally appeared in the superseded American National Standard ANSI PH3.12-1953, have been revised to agree with ANSI PH3.103-1983. These new sizes are consistent with a proposed revision of the International Standard, ISO 1948-1973.

(4) Tapered threads for shutter cable release tip/socket, which originally appeared in the superseded American National Standard ANSI PH3.23-1950, have been revised to agree with ANSI PH3.107-1983 (ANSI/ISO 6053-1979). Release tip threads are interchangeable.

(5) Straight threads for shutter cable release tip/socket, which originally appeared in the American National Standard ANSI PH3.24-1950, have been removed. The ANSI PH3.24-1950 was withdrawn as a standard as of 3 March 1976.

(6) Added lens mount thread data for 16mm and 8mm motion picture cameras.

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SECTION 18 - PHOTOGRAPHIC EQUIPMENT THREADS

1. Scope. This section describes the screw threads commonly used on photographic equipment in order to promote interchangeable connections.

2. Referenced documents.

2.1 Government publications. The issues of the following documents in effect on the date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

Federal standards.

FED-STD-H28/1	Nomenclature, Definitions and Letter Symbols for Screw Threads.
FED-STD-H28/2	Unified Inch Screw Threads - UN and UNR Thread Forms
FED-STD-H28/20	Inspection Methods for Acceptability of UN, UNR, UNJ, M, and MJ Screw Threads
FED-STD-H28/21	Metric Screw Threads

(Activities outside the Federal Government may obtain copies of Federal specifications, standards and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards.

ANSI PH3.101	Photography (cameras) - Tripod connections for American cameras (1/4-inch-20 thread)

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ANSI PH3.102 Photography (cameras)-Tripod connections for heavy-duty or imported cameras (3/8-inch-16 thread with adapter for 1/4-inch-20 tripod screws)
ANSI PH3.103 Photography (cameras)-Lens barrel attachment threads for lens accessories
ANSI PH3.107 Photography (cameras)-Shutter cable release and socket-dimensions
ANSI PH22.76 Threaded Lens Mounts for 16mm and 8mm Motion-Picture Cameras

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. Definitions. Screw thread terms applicable to this standard are defined in FED-STD-H28/1

4. General requirements. Not applicable.

5. Detailed requirements.

5.1 Standard threads. Standard threads for photographic equipment are listed in Table XVIII.1.

5.2 Screw thread acceptability. Inspection methods for acceptability shall be in accordance with FED-STD-H28/20. The required gaging system shall be specified in accordance with that document.1/

MILITARY INTERESTS:

Custodians:

Army - AR
Navy - AS
Air Force - 11

Review Activities:

Army - ER, MI
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

DLA - IS

(DoD Project THDS-0055)

1/ Gaging systems are defined in ANSI B1.3 which is incorporated in FED-STD-H28/20.

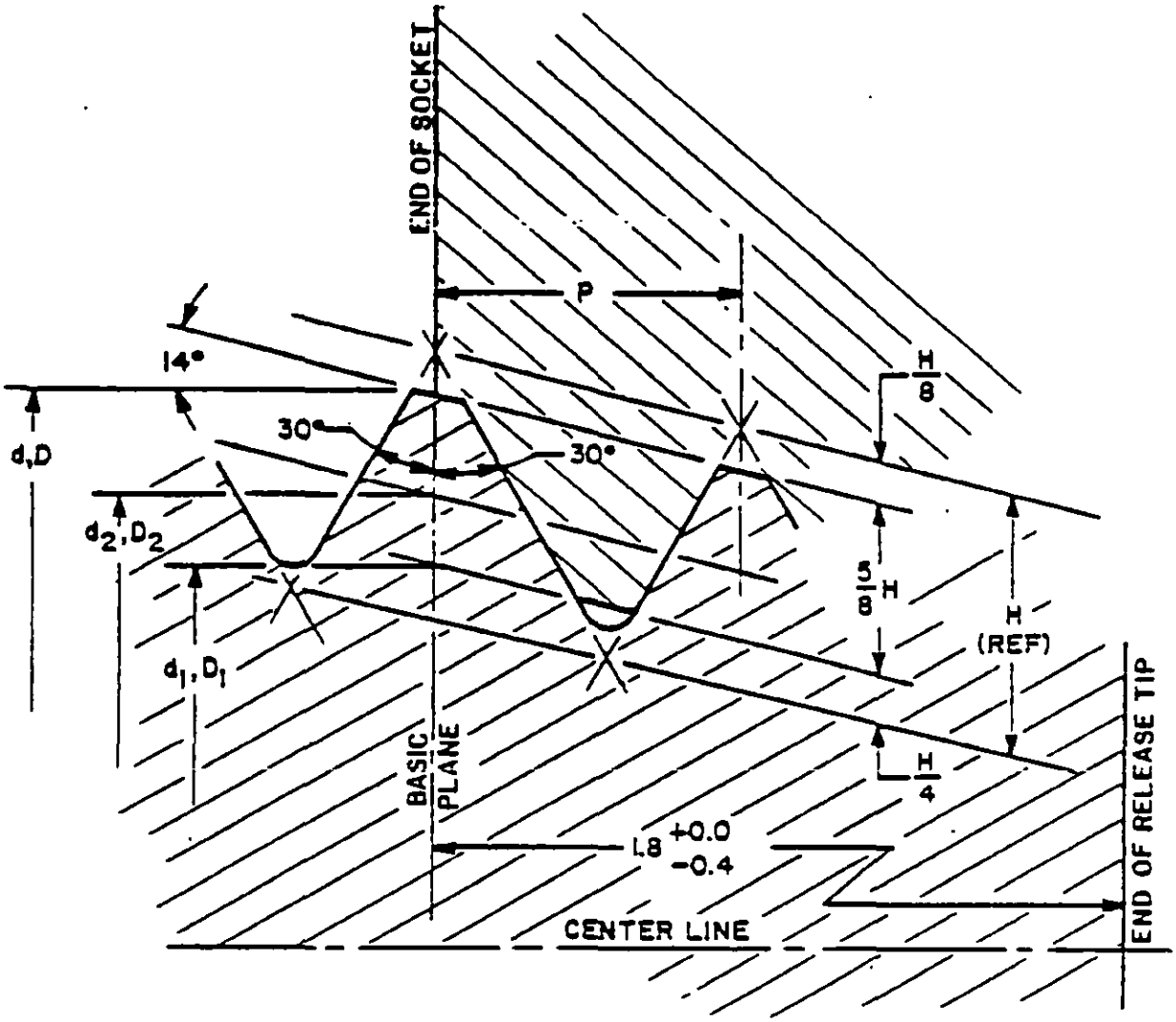
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TABLE XVIII.1. Standard Threads for Photographic Equipment

Connection	Standard Threads	Controlling Thread Standards	Controlling Connection Standards
Tripod Connections	.250-20UNC-1A/1B[1]	FED-STD-H28/2[2]	ANSI PH3.101
	.266-20UNS-1B[3]	FED-STD-H28/2[2]	ANSI PH3.101
	.375-16UNC-1A/1B[4]	FED-STD-H28/2[2]	ANSI PH3.102
Lens Accessory Attachments	M13.5x0.5-6H/6g[5]	FED-STD-H28/21[6]	ANSI PH3.103
	M16.5x0.5-6H/6g[5]		
	M19.5x0.5-6H/6g[5]		
	M22.5x0.5-6H/6g[5]		
	M25.5x0.5-6H/6g[5]		
	M30.5x0.5-6H/6g[5]		
	M35.5x0.5-6H/6g[5]		
	M40.5x0.5-6H/6g[5]		
	M43x0.75-6H/6g[5]		
	M46x0.75-6H/6g[5]		
	M49x0.75-6H/6g[5]		
	M52x0.75-6H/6g[5]		
	M55x0.75-6H/6g[5]		
	M58x0.75-6H/6g[5]		
	M62x0.75-6H/6g[5]		
	M67x0.75-6H/6g[5]		
	M72x0.75-6H/6g[5]		
	M77x0.75-6H/6g[5]		
	M82x0.75-6H/6g[5]		
	M86x1.0-6H/6g[5]		
	M95x1.0-6H/6g[5]	FED-STD-H28/21[6]	ANSI PH.103
Shutter Cable Release Tip/Socket, Tapered Thread Connections	Special See Figure 18.1	ANSI PH3.107	ANSI PH3.107
Shutter Cable Release Tip/Socket, Straight Thread Connections	M3.4x0.5-6H	FED-STD-H28/21[6]	ANSI PH3.107
Lens Mounts for Motion-Picture Cameras	1-32UN-2A(for 16mm)	FED-STD-H28/2[2]	ANSI PH22.76
	.625-32UN-2A(for 8mm)	FED-STD-H28/2[2]	ANSI PH22.76

Notes:
1/ In ISO 1222-1973, an allowance of 0.0043 inch is applied to pitch and major diameters of the internal thread.
2/ The equivalent industry standard for unified screw threads is ANSI B1.1. This standard is incorporated into FED-STD-H28/2.
3/ Non-standard socket thread used to accommodate an oversized tripod screw thread.
4/ In ISO 1222-1973, an allowance of 0.0059 inch is applied to pitch and major diameters of the internal thread.
5/ In ANSI PH3.103, tolerance class is indicated as inch standard class 2.
6/ The equivalent industry standard for metric M profile screw threads is ANSI/ASME B1.13M. This standard is incorporated into FED-STD-H28/21.

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BASIC PLANE DIMENSIONS	
MAJOR DIAMETER d,D	=3.400
PITCH DIAMETER d_2,D_2	=3.082
MINOR DIAMETER d_1,D_1	=2.870

THREAD FORM DIMENSIONS	
PITCH	=0.5
FUNDAMENTAL TRIANGLE HEIGHT, H	=0.424
TRUNCATION AT MAJOR DIAMETER, H/8	=0.053
TRUNCATION AT MINOR DIAMETER, H/4	=0.106
THREAD HEIGHT, 5/8 H	=0.265

FIGURE 18.1 THREADS FOR SHUTTER CABLE RELEASE TIP (EXTERNAL TAPER THREAD) AND SOCKET (INTERNAL TAPER THREAD)
NOTE: All dimensions are in millimeters unless indicated otherwise.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
FED-STD-H/18	PHOTOGRAPHIC EQUIPMENT THREADS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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